UNIT 2

# Crack Growth, Fracture Mechanisms, Griffith Theory & Energy Release Rate

*fracture mechanisms | theoretical strength | Inglis | Griffith | G, R-curve, compliance*

## I. Fracture – Basics

**Fracture** = separation of a body into two or more pieces under stress. It has two stages: (i) **crack initiation** (nucleation) and (ii) **crack propagation**. Types of fracture:

| Basis | Types |
|---|---|
| Plastic deformation | Brittle / Ductile / Quasi-brittle |
| Crack path | Transgranular (through grains) / Intergranular (along grain boundaries) |
| Loading | Static overload, fatigue, creep, impact, stress-corrosion, hydrogen-assisted |
| Crack speed | Stable (slow, needs rising load) / Unstable (fast, self-driven) |

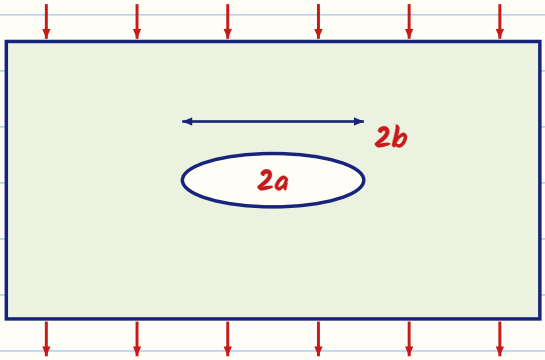
### I.1 Theoretical cohesive strength

Atoms separated by a distance x from equilibrium spacing $a_0$: stress-separation curve approximated by a sine:

$$\sigma = \sigma_{th} \sin(2\pi x/\lambda) \rightarrow \sigma_{th} = \sqrt{(E\,\gamma_s / a_0)} \approx E/10 \text{ (Orowan)}$$

- Derivation: work to separate two planes = $2\gamma_s = \int \sigma\, dx = \sigma_{th}\, \lambda/\pi$ ; for small x, Hooke $\sigma = E\, x/a_0 = \sigma_{th}\, (2\pi x/\lambda) \Rightarrow \lambda = 2\pi\, a_0\, \sigma_{th} / E$ ; then $2\gamma_s = \sigma_{th}\, \lambda/\pi = 2\, a_0\, \sigma_{th}^2/E \Rightarrow$ **$\sigma_{th} = \sqrt{(E\gamma_s/a_0)}$**.
- For steel E = 210 GPa → $\sigma_{th}$ ≈ 20-30 GPa, but measured fracture strength ≈ 0.2-2 GPa, i.e. 10-100 times lower! Glass: E = 70 GPa → $\sigma_{th}$ ≈ 7-10 GPa; measured ≈ 70 MPa.
- Griffith (1920) explained the discrepancy by **pre-existing microscopic cracks** which amplify stress locally.

## 2. Inglis Stress Concentration




**Inglis (1913):**

$\sigma_{max} = \sigma(1 + 2a/b)$

$\rho = b^2/a \rightarrow \sigma_{max} = \sigma(1+2\sqrt{(a/\rho)})$

**sharp crack: $\rho \rightarrow 0$ , $\sigma_{max} \rightarrow \infty$**

(so stress criterion fails!)

Fig 2.1 Elliptical hole in an infinite plate (Inglis 1913)

- Max stress at the end of the major axis: $\sigma_{max} = \sigma\ (1 + 2a/b)$. With root radius $\rho = b^2/a$: $\sigma_{max} = \sigma\ (1 + 2\sqrt{(a/\rho)})$.
- Stress concentration factor $K_t = \sigma_{max}/\sigma = 1 + 2a/b$. Circle ($a = b$): $K_t = 3$.
- For a thin slit/sharp crack, $b \rightarrow 0$ ($\rho \rightarrow 0$) and $\sigma_{max} \rightarrow \infty$ : **a stress criterion cannot be used.** Need an energy (Griffith) or K approach.
- For $a/\rho >> 1$: $\sigma_{max} \approx 2\sigma\sqrt{(a/\rho)}$. Real crack tips have $\rho \approx$ atomic radius $\Rightarrow \sigma_{max} \approx \sigma_{th} \Rightarrow$ fracture at $\sigma_f = \sigma_{th}\ \sqrt{(\rho/4a)}$ — which actually matches Griffith's result when $\rho \approx a_0$.

## 3. Griffith's Theory of Fracture (1921)

**Idea:** a crack can grow only if the total energy of the system **decreases** (or stays constant) – the energy released from strain energy must be at least the energy needed to create the new surfaces. (First law of thermodynamics: balance.)

**Assumptions of Griffith theory**

> Material is perfectly elastic, homogeneous, isotropic (ideally brittle, e.g. glass).

> Pre-existing sharp crack of length 2a, through-thickness, in an infinitely wide plate (B = thickness).

> Energy needed for crack growth is only the **surface energy** ($\gamma_s$ per unit area, two surfaces).

> Crack grows quasi-statically; kinetic energy neglected; applied stress $\sigma$ is constant & uniform far away.

### 3.1 Derivation (energy balance)

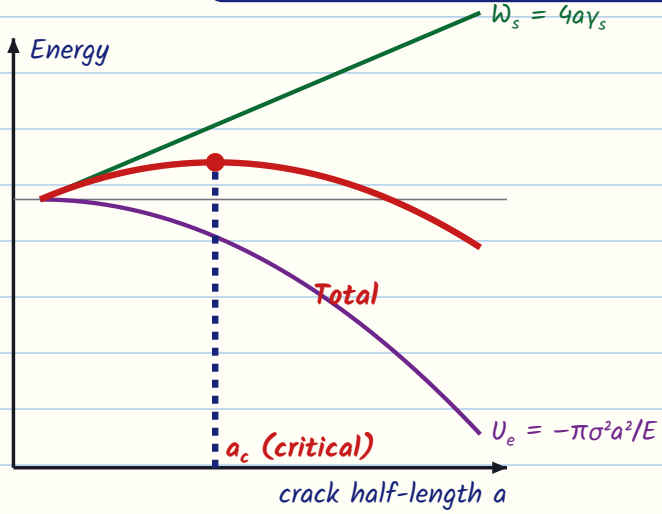
Total energy of the cracked plate: $U = U_0 + U_a + U_\gamma - F$

- $U_0$ = elastic energy of uncracked plate (constant).
- $U_a$ = change in elastic energy due to crack (Inglis solution) = $-\pi\ \sigma^2\ a^2\ B\ /\ E$ (plane

stress).

- $U_\gamma$ = surface energy = (2 faces × 2a × B) $\gamma_s$ = 4 a B $\gamma_s$.
- F = work done by external forces; for **fixed grips** F = 0 (no displacement); for fixed load, F = $-2U_a$-type term; result for G is same.

$$U = U_0 - \pi \sigma^2 a^2 B / E + 4 a B \gamma_s$$




**$a > a_c$ : UNSTABLE**
energy released > needed
→ crack runs

**$a < a_c$ : STABLE**
more energy needed to grow

$d(U_e + W_s)/da = 0$
$\Rightarrow \sigma_f = \sqrt{(2E\gamma_s/\pi a)}$

Fig 2.2 Griffith energy balance – the energy barrier at critical crack size

Equilibrium (crack at point of instability): **dU/da = 0**

$$-2\pi \sigma^2 a B / E + 4 B \gamma_s = 0$$

$$\sigma_f = \sqrt{( 2 E \gamma_s / (\pi a) )}$$ plane stress

Written as $\sigma_f \sqrt{(\pi a)} = \sqrt{(2E\gamma_s)}$ = constant. i.e. for a given material, strength × √(crack size) = const. Here crack size = 2a (central crack); for an edge crack of length a, the same formula applies with a = the edge-crack length (up to a free-surface factor 1.12).

**Plane strain form**

> Replace E by $E' = E/(1-\nu^2)$: $\sigma_f = \sqrt{( 2 E \gamma_s / (\pi (1-\nu^2) a) )}$
> Critical crack length: $a_c = 2 E \gamma_s / (\pi \sigma^2)$ (plane stress)
> Condition for growth: $G \geq 2\gamma_s$ where $G = \pi \sigma^2 a / E$.

## 3.2 Modifications – Irwin & Orowan

- Griffith works only for ideally brittle solids (glass, ceramics, mica). In metals, plastic work $\gamma_p$ at the crack tip is ~ $10^3$ × $\gamma_s$ (e.g. steel $\gamma_s \approx 1$ J/m², $\gamma_p \approx 10^3$–$10^4$ J/m²).
- **Irwin–Orowan modification:** replace $\gamma_s$ by $(\gamma_s + \gamma_p)$:

$$\sigma_f = \sqrt{( 2E(\gamma_s + \gamma_p) / (\pi a) )} \approx \sqrt{(2E\gamma_p/\pi a)}$$

- Irwin defined $G_c = 2(\gamma_s + \gamma_p)$ = critical energy release rate (fracture toughness in energy terms); fracture when $G = G_c$.

### Worked Example 1 (glass)

A glass plate: $E = 70$ GPa, $\gamma_s = 0.5$ J/m², contains a central crack $2a = 2$ μm. Find fracture stress (plane stress).

$a = 1\times10^{-6}$ m. $\sigma_f = \sqrt{(2\ E\ \gamma_s\ /(\pi\ a))} = \sqrt{(2 \times 70\times10^{9} \times 0.5\ /(\pi \times 10^{-6}))} = \sqrt{(2.228\times10^{16})} =$ **$1.49 \times 10^{8}$ Pa ≈ 149 MPa**.

Check: 1 mm crack ($a = 0.5$ mm): $\sigma_f = \sqrt{(7\times10^{10}/(\pi\times5\times10^{-4}))} =$ **6.7 MPa** – shows how strength falls with flaw size.

### Worked Example 2 (steel)

Large steel plate, $E = 210$ GPa, $G_c = 2(\gamma_s+\gamma_p) = 20$ kJ/m² . Applied $\sigma = 300$ MPa. Critical half crack length?

From $G = \pi\ \sigma^2\ a\ /\ E = G_c$ : $a_c = G_c\ E\ /(\pi\ \sigma^2) = 20000 \times 210\times10^{9}\ /(\pi \times 9\times10^{16}) =$ **0.01485 m ≈ 14.9 mm** (crack length $2a_c \approx 29.7$ mm).

Equivalent $K_c = \sqrt{(E\ G_c)} = \sqrt{(210\times10^{9} \times 20000)} = 64.8$ MPa√m.

## 4. Energy Release Rate (G)

**Definition (Irwin 1956):** $G$ = energy available per unit increase in crack area (per unit thickness per unit crack length). It is the **crack driving force**.

$$G = -\ d\Pi/dA = -(1/B)\ d\Pi/da$$

$\Pi$ = potential energy = $U - F$

- Units: J/m² = N/m. ('Release rate' is not a time rate.)
- For a through crack in infinite plate: $G = \pi\ \sigma^2\ a\ /\ E'$ ($E'$ = $E$ plane stress, $E/(1-\nu^2)$ plane strain).
- Crack extends when $G \geq R$ where $R$ = crack resistance (energy absorbed per unit crack area) = $G_c$ for a flat R-curve. For Griffith, $R = 2\gamma_s$.

### 4.1 Fixed grip vs fixed load

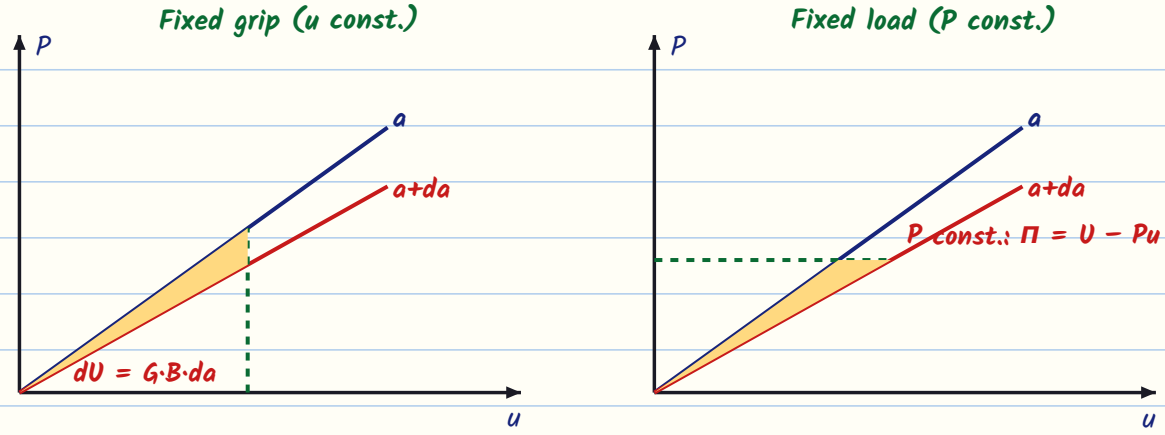


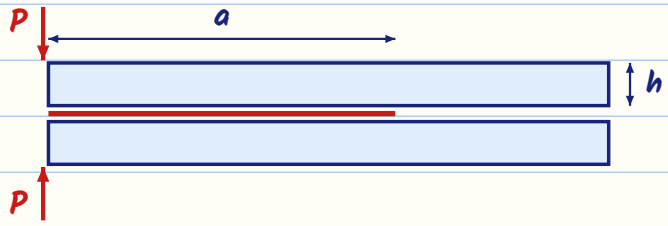

Fig 2.3 Load–displacement response for crack length a and a+da

- **Fixed grip (u const):** crack growth lowers load P; the elastic energy $U = \tfrac{1}{2}Pu$ decreases; $U = u^2/(2C) \Rightarrow G = -(1/B)(\partial U/\partial a)_u = (u^2/2BC^2)(dC/da) = (P^2/2B)(dC/da)$, with compliance $C = u/P$.
- **Fixed load (P const):** U increases by $\tfrac{1}{2}P\,du$ but external work $P\,du$ is done; $\Pi = U - Pu = -\tfrac{1}{2}Pu$ so $G = (1/B)(\partial(\tfrac{1}{2}P\,u)/\partial a)_P = (P^2/2B)(dC/da)$.
- Both give the **same G**:

$$G = (P^2 / 2B) \cdot (dC/da)$$ compliance $C = u/P$

Compliance method is used experimentally: measure C vs a for specimens with different crack lengths, differentiate numerically, get G (and K).

## 4.2 Example – Double Cantilever Beam (DCB)




**Double Cantilever Beam (DCB)**

$\delta = 8Pa^3/(E\,B\,h^3)$

$C = \delta/P = 8a^3/(E\,B\,h^3)$

$G = (P^2/2B)(dC/da)$

$G = 12P^2a^2/(E\,B^2\,h^3)$

Fig 2.4 DCB specimen (each arm is a cantilever of depth h)

Each arm: deflection $P\,a^3/(3EI)$ with $I = B\,h^3/12$ → total opening $\delta = 8\,P\,a^3/(E\,B\,h^3)$. $C = 8a^3/(E\,B\,h^3)$. $dC/da = 24\,a^2/(E\,B\,h^3)$.

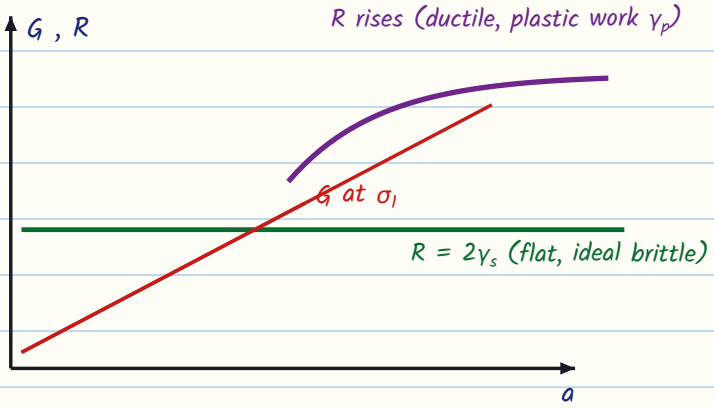
$G = (P^2/2B)(24a^2/(E\,B\,h^3)) = 12\,P^2\,a^2\,/\,(E\,B^2\,h^3)$. Note $G \propto a^2$, so G increases with a at constant P (unstable at fixed load); for fixed displacement δ: $G = 3Eh^3\delta^2/(16a^4)$ decreases with a → **stable** crack growth. (Used in adhesive and composite delamination tests, $G_{Ic}$.)

### Worked Example 3 (DCB)

DCB: $E = 70$ GPa, $h = 10$ mm, $B = 20$ mm, $a = 80$ mm. Find $P$ at which $G = G_c = 400$ J/m².

$P^2 = G\,E\,B^2\,h^3 / (12\,a^2) = 400 \times 70\times10^9 \times (0.02)^2 \times (0.01)^3 / (12 \times (0.08)^2) =$ $400\times70e9\times4e-4\times1e-6/(0.0768) = 1.12\times10^4/0.0768 = 1.458\times10^5 \rightarrow$ **$P \approx 382$ N.**

## 5. R-curve and Stability of Crack Growth




**Stability condition**

Growth starts : $G = R$
Stable growth : $dG/da < dR/da$
Unstable : $dG/da \geq dR/da$

Fig 2.5 Crack-driving force G and resistance R

- **Flat R-curve** (ideally brittle; plane strain thick plates): $R = G_c$ constant. Growth starts at $G = R$ and is immediately unstable at fixed load since $dG/da > 0 = dR/da$.
- **Rising R-curve** (ductile; plane stress thin sheets): R increases with $\Delta a$ because the plastic zone grows. Stable tearing occurs till tangency of G-line with R-curve; instability when **$G = R$ and $dG/da = dR/da$**.
- Stable growth: $dG/da < dR/da$. Used in ASTM ES61 (K-R curve) for thin sheet aluminium alloys.

### Relation to Griffith

For Griffith's plate, $G = \pi\,\sigma^2\,a/E \rightarrow$ increases linearly with a at fixed $\sigma$. Instability when $G = 2\gamma_s$. For a given crack, increasing $\sigma$ raises the G-line.

## 6. Crack Growth and Fracture Mechanisms

### 6.1 Ductile fracture – micro-void coalescence

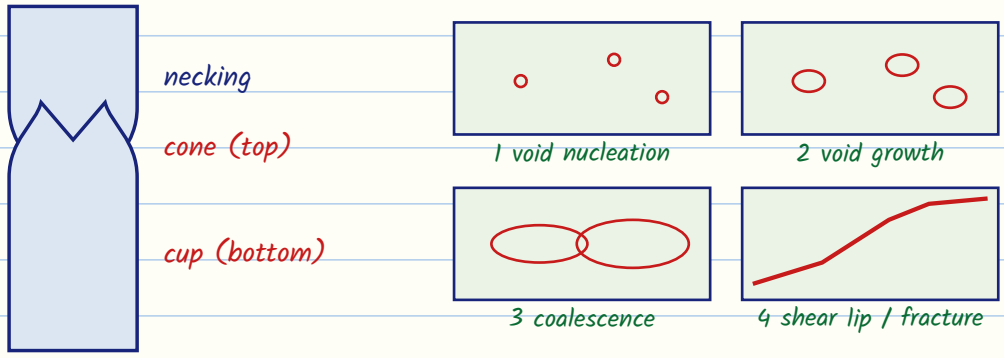



Fig 2.6 Cup-and-cone fracture and its microscopic stages

1. **Necking** begins at UTS: triaxial stress develops at the neck centre.
2. **Void nucleation** at second-phase particles / inclusions (decohesion or cracking).
3. **Void growth** under hydrostatic tension; then **coalescence** by internal necking of ligaments → central flat crack.
4. Crack spreads outward; the outer ring fails by **shear at 45°** (shear lip) → **cup and cone**. Surface shows **dimples**.

## 6.2 Brittle fracture – cleavage

- **Cleavage**: transgranular separation along low-index crystallographic planes (e.g. {100} in BCC iron) with very little plasticity. Occurs in BCC & HCP metals at low T, high strain rate, triaxial stress; ceramics, glass.
- Fractograph: flat facets; **river patterns** (steps merging in the direction of crack growth, 'tributaries' join 'river'); **chevron (herringbone) marks** point back to the origin; **tongues** from twins.
- Cleavage strength follows a Hall-Petch type relation $\sigma_f \propto d^{-1/2}$ ($d$ = grain size) → fine grain gives higher toughness.

## 6.3 Intergranular fracture

- Crack follows grain boundaries: segregation of impurities (temper embrittlement – P, Sn, Sb), hydrogen embrittlement, stress-corrosion cracking, creep cavitation, grain-boundary carbides. Surface looks like 'rock candy'.

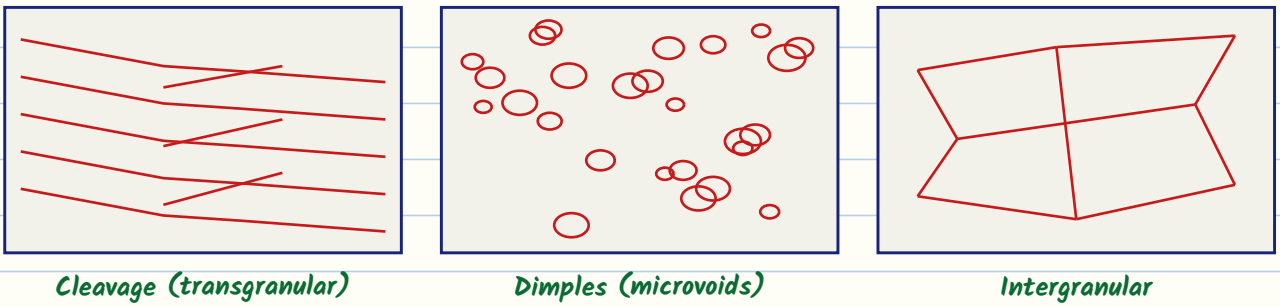



Fig 2.7 Fracture surfaces: cleavage / dimples / intergranular

## 6.4 Quasi-cleavage & fatigue

- **Quasi-cleavage** : mix of cleavage facets and tear ridges (tempered martensite).
- **Fatigue**: striations (Stage II), beach marks. Mechanism: plastic blunting & re-sharpening of crack tip each cycle (Laird model): tensile half – crack opens & blunts by slip; compressive half – closes, re-sharpens → one striation per cycle.

| Mechanism | Typical conditions | Fracture surface | Energy |
|---|---|---|---|
| Cleavage | BCC, low T, high strain-rate, thick | Flat, shiny facets, river marks | Low |
| Micro-void coalescence | FCC, high T, low rate | Dimples, fibrous, dull | High |
| Intergranular | Embrittled boundaries, SCC, creep | Faceted grains | Low-med |
| Fatigue | Cyclic loading | Striations, beach marks | — |
| Shear rupture | Very pure/soft metals | Chisel point, 100 % necking | High |

## 6.5 Factors that influence fracture

- **Temperature** (DBTT); **strain rate** (higher → brittle); **triaxiality/constraint** (notch, thick section → brittle).
- **Thickness**: $K_c$ falls from the plane-stress value to the plane-strain $K_{Ic}$ as $B \uparrow$; shear lips shrink.
- **Grain size** (fine = tougher), **inclusions**, **heat treatment**, **irradiation**, **environment** (hydrogen, SCC), **residual stress** & **weld defects**.
- **Strength level**: for a given alloy, $K_{Ic}$ usually decreases as $\sigma_{ys}$ increases (trade-off).

**Thickness requirement & plane strain (ASTM E399)**

> $B \geq 2.5\,(K_{Ic}/\sigma_{ys})^2$ and $a \geq 2.5\,(K_{Ic}/\sigma_{ys})^2$ and $(W-a) \geq 2.5\,(K_{Ic}/\sigma_{ys})^2$.

> Example: $K_{Ic} = 60$ MPa$\sqrt{m}$, $\sigma_{ys} = 500$ MPa $\Rightarrow B \geq 2.5\,(0.12)^2 = 0.036$ m = 36 mm.

# 6A. Additional Topics

## A. Griffith's experiments & size effect

- Griffith drew glass fibres: strength rose as diameter fell – about **170 MPa** for bulk glass, rising to ~**3.5 GPa** for fibres ≈ 2.5 μm thick (near $\sigma_{th}$). Smaller specimens can only contain smaller flaws ⇒ confirms $\sigma_f \propto 1/\sqrt{a}$.
- Hence **whiskers** and glass/carbon fibres are very strong; large brittle components are weak. Statistical (Weibull) strength depends on volume: weakest-link theory.

## B. Why surface energy alone is insufficient for metals

| Material | $\gamma_s$ (J/m²) | $G_c = 2(\gamma_s+\gamma_p)$ (J/m²) |
|---|---|---|
| Glass | ≈ 0.5 – 1 | ≈ 2 – 10 |
| Alumina (ceramic) | ≈ 1 | ≈ 20 – 40 |
| Mild/structural steel | ≈ 1-2 | ≈ $10^4$ – $10^5$ |
| Aluminium alloy | ≈ 1 | ≈ $10^4$ – $3\times10^4$ |

## C. Dynamic fracture & crack arrest (Mott)

- Once G exceeds R, extra energy (G – R) goes into **kinetic energy** of material near the moving crack; crack accelerates.
- **Mott / Roberts-Wells:** terminal crack speed ≈ **0.38 √(E/ρ)** (about 0.2-0.4 × shear-wave speed). Steel: ≈ 0.38 × √(210×10⁹/7850) ≈ **1960 m/s**; practical speeds 1000-2000 m/s. Branching occurs near the limiting speed (glass shatters).
- For dynamic loading $K_I(t) \rightarrow K_{Id}$ (dynamic toughness) ≤ $K_{Ic}$ with increased strain rate; **crack-arrest toughness $K_{Ia}$** < $K_{Ic}$ is the K below which a running crack stops.

## D. Sub-critical (environment-assisted) crack growth

- Cracks can grow at $K_I < K_{Ic}$ with time: **stress-corrosion cracking (SCC)**, hydrogen-induced cracking, creep crack growth, corrosion fatigue.
- Threshold **$K_{Iscc}$** : below it no growth in that environment (e.g. high-strength steel in sea water: $K_{Iscc}$ ≈ 0.2-0.4 $K_{Ic}$). Growth rate da/dt vs K shows 3 regions: (I) steep rise, (II) plateau (diffusion controlled), (III) $K \rightarrow K_{Ic}$.
- Design rule for sustained loads: keep $K_I < K_{Iscc}$. Time to failure from $K_i$ to $K_{Ic}$ is found by integrating $da/dt = f(K)$.

## E. Fractography (failure analysis tools)

- **Macro-fractography** (eye, stereo-microscope): origin, beach marks, chevron, shear lips, direction of crack travel.
- **SEM** (micro-fractography): dimples, facets, striations (spacing → da/dN), intergranular facets; EDS for corrosion products. **TEM** replicas for fine striations.
- **Failure-analysis steps:** collect history → preserve fracture faces → visual → NDT → chemical/ mechanical tests → fractography → stress analysis (FM) → report & remedy.

## Solved Numerical Problems

**P1 (Inglis).** Elliptical hole a = 5 mm, b = 0.5 mm in plate with σ = 100 MPa. $\sigma_{max}$ = 100 (1 + 2×5/0.5) = **2100 MPa** ($K_t$ = 21); root radius ρ = $b^2/a$ = 0.05 mm.

**P2 (ceramic).** Alumina, E = 380 GPa, $\gamma_s$ = 1 J/m². For σ = 100 MPa the critical half-crack length $a_c = 2E\gamma_s/(\pi\sigma^2) = 2\times380\times10^9\times1/(\pi\times10^{16}) =$ **$2.4\times10^{-5}$ m = 24 µm** (so grain-size

flaws control strength).

**P3 (plane strain).** Steel $E = 210$ GPa, $\nu = 0.3$, $G_c = 10$ kJ/m², $a = 10$ mm. Plane stress: $\sigma_f = \sqrt{(E\,G_c/(\pi\,a))} =$ **258 MPa**; plane strain: $\sigma_f = \sqrt{(E\,G_c\,/(\pi(1-\nu^2)a))} =$ **271 MPa** (the plane-strain value is higher for the same $G_c$, but real $G_c$ is lower in plane strain).

**P4 (compliance).** At $P = 2$ kN, $B = 10$ mm, compliance rises from $C = 1.00\times10^{-7}$ to $1.10\times10^{-7}$ m/N when a goes 50 → 52 mm. $dC/da \approx 5\times10^{-5}$ (m/N)/m. $G = (P^2/2B)(dC/da) = (4\times10^6/0.02)\times5\times10^{-5} =$ **$1.0\times10^4$ J/m²**; $K = \sqrt{(E\,G)} =$ **45.8 MPa√m** ($E = 210$ GPa, plane stress).

**P5 (thickness).** Alloy $\sigma_{ys} = 450$ MPa, $K_{Ic} = 80$ MPa√m: $B \geq 2.5\,(80/450)^2 = 0.079$ m = **79 mm** → a thin sheet will not give a valid $K_{Ic}$; use $K_c$ / R-curve or EPFM.

## 6B. G for Standard Cases & Summary

| Configuration | Energy release rate G |
|---|---|
| Central crack 2a, infinite plate, σ | $\pi\,\sigma^2\,a\,/\,E'$ |
| Single edge crack a, semi-infinite plate | $1.12^2\,\pi\,\sigma^2\,a\,/\,E' \approx 1.25\,\pi\,\sigma^2\,a/E'$ |
| DCB (arm depth h), load P | $12\,P^2\,a^2\,/\,(E\,B^2\,h^3)$ |
| Infinite strip (height 2h), fixed grip, strain ε | $G = \sigma^2\,h\,/\,E = E\,\varepsilon^2\,h$ (independent of a) |
| Penny-shaped crack radius a, σ | $4\,(1-\nu^2)\,\sigma^2\,a\,/\,(\pi\,E)$ |
| Any body (compliance method) | $(P^2/2B)\,dC/da$ |

### Griffith vs Irwin vs Orowan (revision)

> **Griffith (1921):** energy balance, only surface energy, ideally brittle: $\sigma_f = \sqrt{(2E\gamma_s/\pi a)}$.

> **Irwin (1948):** includes plastic work: $G_c = 2(\gamma_s+\gamma_p)$; introduced $G = -d\Pi/dA$ (1956) and K (1957); $G = K^2/E'$.

> **Orowan (1949):** independently suggested $\gamma_p \gg \gamma_s$ for metals; $\sigma_f = \sqrt{(2E(\gamma_s+\gamma_p)/\pi a)}$.

### Worked Example 4 (strip, fixed grip)

Thin strip of height $2h = 100$ mm, $E = 70$ GPa, $G_c = 20$ kJ/m². Gripped at fixed displacement. Find the stress/strain at which a long edge crack runs.

$G = \sigma^2\,h\,/\,E = G_c \Rightarrow \sigma_c = \sqrt{(G_c\,E\,/\,h)} = \sqrt{(2\times10^4 \times 7\times10^{10}\,/\,0.05)} =$ **167 MPa** ; $\varepsilon_c = \sigma_c/E =$ **0.0024** (0.24 %).

G is independent of crack length ⇒ once started, the crack grows at constant G (steady-state propagation).

## Worked Example 5 (penny-shaped crack)

Buried penny-shaped crack, radius $a = 5$ mm, $\sigma = 150$ MPa, steel ($E = 210$ GPa, $\nu = 0.3$). $K_I = 2\sigma\sqrt{(a/\pi)} = 2\times150\times\sqrt{(0.005/\pi)} =$ **12.0 MPa√m**. $G = (1-\nu^2)K^2/E = 0.91\times(11.97\times10^6)^2/(210\times10^9) =$ **621 J/m²**.

## Key Points to Remember

- A crack grows when the **rate of energy release G ≥ rate of energy absorption R** (necessary condition – Griffith criterion is necessary, not sufficient, since the energy must also be transferable at the tip by stress).
- G is a **global energy** quantity; K is a **local stress-field** quantity; they are linked by $G = K^2/E'$ (Unit III).
- G depends on boundary conditions only through the geometry/compliance, **not** on whether grips are fixed or the load is fixed.
- Fixed-grip loading generally gives more stable crack growth (G falls as a grows for a DCB); dead-weight loading is the most unstable.

## Unsolved Practice Problems (with answers)

1. Glass, $E = 70$ GPa, $\gamma_s = 0.6$ J/m²; central crack of length $2a = 0.2$ mm. Find $\sigma_f$. Ans: **≈ 16.4 MPa**
2. Steel, $E = 210$ GPa, $G_c = 15$ kJ/m², $\sigma = 200$ MPa. Find $a_c$. Ans: **≈ 25.1 mm**
3. DCB: $P = 300$ N, $a = 60$ mm, $h = 8$ mm, $B = 15$ mm, $E = 70$ GPa. Find G and K. Ans: **G ≈ 482 J/m²**, $K = \sqrt{(EG)} \approx$ **5.8 MPa√m**
4. Show that for Griffith's plate $\sigma_f^2\, a =$ const. and plot $\sigma_f$ vs a (hyperbola-type curve; log–log slope = −1/2).

## 2-mark Short Answers

**Q. Define surface energy.** Energy required to create unit area of new free surface (J/m²).

**Q. State Griffith criterion.** A crack propagates if the elastic energy released by an increment of growth ≥ the surface energy required.

**Q. What is compliance?** C = displacement/load = u/P; reciprocal of stiffness; depends on crack length.

**Q. Difference between G and K?** G = energy approach (global), K = stress-field approach (local); $G = K^2/E'$.

**Q. What is an R-curve?** Plot of crack resistance R vs crack extension Δa; flat for brittle, rising for ductile materials.

**Q. Give two features of cleavage.** Flat bright facets, river patterns, low energy, along crystal planes.

**Q. What is a dimple rupture?** Fracture by void nucleation, growth and coalescence –

cup-and-cone surfaces have dimples.

## 7. Exam Questions & Quick Answers

1. Derive Griffith's criterion for brittle fracture. State the assumptions and the modification for ductile materials.
2. Define energy release rate. Show that $G = (P^2/2B)\, dC/da$ and that it is the same for fixed load and fixed grip.
3. Explain the R-curve and conditions for stable and unstable crack growth.
4. Describe the mechanisms of ductile and brittle fracture with sketches.
5. Why does the Inglis solution fail to give a fracture criterion for sharp cracks?
6. Compute G and critical load for a DCB specimen (numerical).

**Formula sheet – Unit II**

> $\sigma_{th} = \sqrt{(E\gamma_s/a_0)} \approx E/10$ | $\sigma_{max} = \sigma(1+2\sqrt{(a/\rho)})$

> Griffith: $\sigma_f = \sqrt{(2E\gamma_s/\pi a)}$ ; plane strain $E \to E/(1-\nu^2)$ ; ductile: $\gamma_s \to \gamma_s+\gamma_p$

> $G = \pi\sigma^2 a/E'$ ; $G = (P^2/2B)(dC/da)$ ; $G_c = R = 2(\gamma_s+\gamma_p)$

> DCB: $G = 12P^2a^2/(E\, B^2\, h^3)$ ; Instability: $G = R$ and $dG/da \geq dR/da$